HEART RATE MONITORING SYSTEM USING ARDUINO

A Smart System for Real-Time Heart Rate Monitoring

PROJECT PRESENTATION

Fida

Intoduction

- Heart rate monitoring is crucial for detecting health issues and ensuring well-being.
- This project uses Arduino to measure heart rate and provide visual feedback based on BPM (Beats Per Minute).
- A green light indicates a normal heart rate, while a red light signals an abnormal heart rate.
- The real-time BPM reading is displayed on an LED display for easy monitoring.

Components Used

- **Arduino Uno** – The microcontroller that processes the heart rate data.
- **Pulse Sensor** – Detects the user's heartbeat and sends data to Arduino.
- **LEDs (Red & Green)** – Provide visual feedback based on heart rate.
- **7-Segment LED Display (or LCD Display)** – Shows the heart rate in BPM.
- **Resistors and Jumper Wires** – For circuit connections and stability.
- **Breadboard** – For assembling the circuit without soldering.

Working Principle

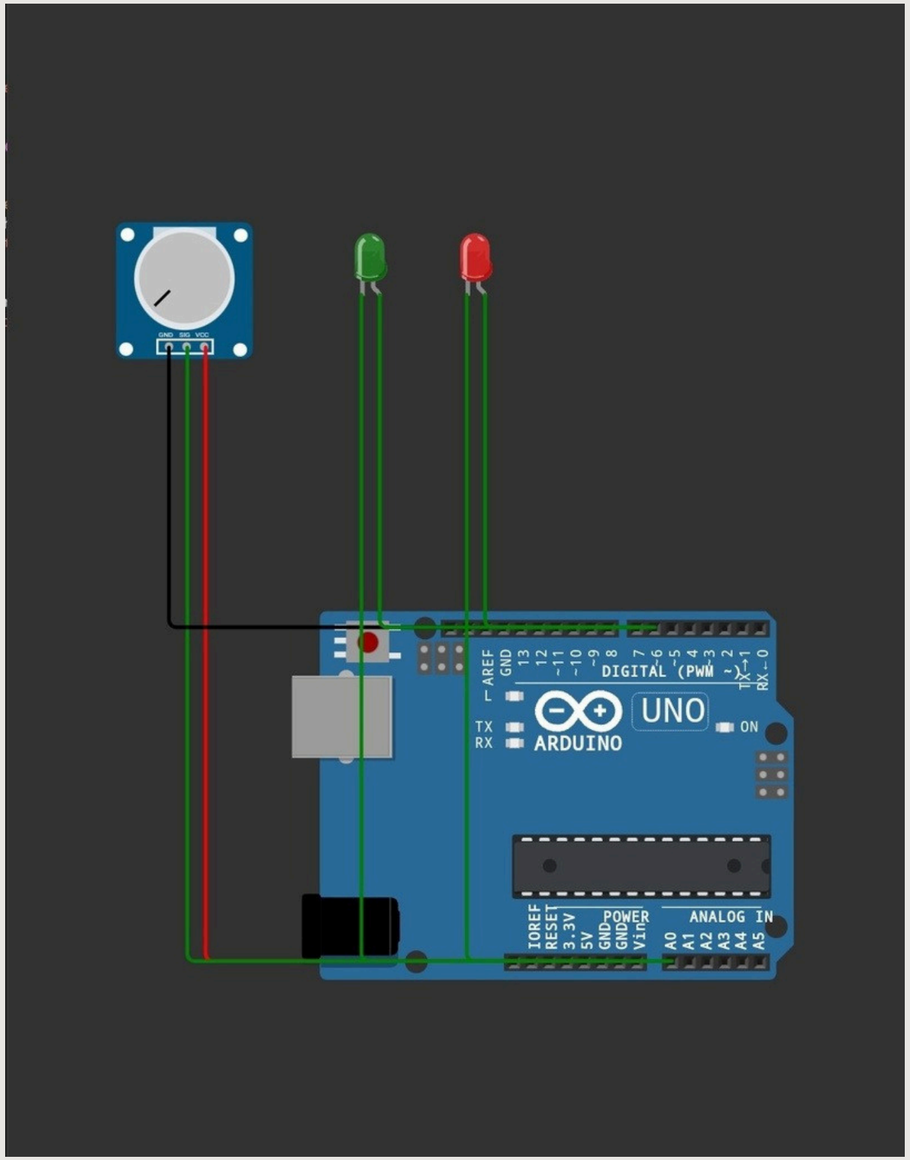
1. The **pulse sensor** detects heartbeat by sensing blood flow changes in the fingertip.
2. This data is sent to the Arduino, which calculates the **BPM (Beats Per Minute).**
3. Based on predefined thresholds:

- If the BPM is within the **normal range (60–100 BPM)**, the **green LED lights up.**
- If the BPM is **too low (<60 BPM) or too high (>100 BPM)**, the **red LED lights up.**

4. The BPM value is **displayed in real time** on the LED display.

Circuit Diagram

- **Pulse Sensor → Arduino Analog Input Pin-A0**
- **Green LED → Digital Output Pin-7**
- **Red LED → Digital Output Pin-6**
- **LED Display**



Arduino Code

```
#define SENSOR_PIN A0
#define GREEN_LED 7
#define RED_LED 6

void setup() {
  pinMode(GREEN_LED, OUTPUT);
    pinMode(RED_LED, OUTPUT);
      Serial.begin(9600);
      }

      void loop() {
        int sensorValue = analogRead(SENSOR_PIN);


              int heartRate = map(sensorValue, 0, 1023, 40, 120);


                  Serial.print("Heart Rate: ");
                    Serial.print(heartRate);
                      Serial.println(" BPM");


                          if (heartRate >= 60 && heartRate <= 100) {
                              digitalWrite(GREEN_LED, HIGH);
                                  digitalWrite(RED_LED, LOW);
                                    } else {
                                        digitalWrite(GREEN_LED, LOW);
                                            digitalWrite(RED_LED, HIGH);
                                              }

                                                delay(1000);
                                                }


```

Advantages of the System

- **Real-time Monitoring:** Instantly displays heart rate data.
- **Easy to Use:** Simple setup with visual indicators.
- **Portable:** Small and lightweight for personal health tracking.
- **Cost-Effective:** Uses inexpensive components.
- **Customizable:** Can be modified for different BPM ranges or alerts.

Applications

- **Personal Health Monitoring** – Helps individuals track their heart rate.
- **Fitness Tracking** – Useful for athletes to monitor their heart rate during exercise.
- **Medical Applications** – Can assist doctors in basic heart rate assessments.
- **Educational Purposes** – A great project for learning about biomedical electronics.

Limitations & Future Improvements

1. **Limitations:**

- May not be as accurate as medical-grade devices.
- Affected by movement or improper sensor placement.

2.**Future Improvements:**

- Adding a buzzer alarm for critical heart rate conditions.
- Wireless data transmission via Bluetooth or Wi-Fi.
- Integrating with a mobile app for remote monitoring.

Conclusion

- The **Heart Rate Monitoring System using Arduino** is a simple yet effective project for real-time health tracking.
- It provides **visual alerts and BPM display**, making it user-friendly.
- Future improvements can enhance its accuracy and usability in **medical and fitness applications**.

THANK YOU!